

From Quantum Seeds to the Cosmic Web

Lecture notes for the 10th Vietnam School of Astrophysics (VSOA10)

Two lectures of 90 minutes each, with two hands-on sessions

Nhat-Minh Nguyen

ICISE, Quy Nhon, Vietnam

27 July – 1 August 2026

Abstract

These notes follow a single path: from the primordial fluctuations left by inflation to the cosmic web traced by galaxy surveys. Lecture 1 builds the linear matter power spectrum, starting from the Newtonian equations for a self-gravitating fluid and the four assumptions they rest on, then linear growth, Gaussian random fields, and the way horizon crossing during radiation domination carves the turnover into $P_L(k)$. Lecture 2 takes that spectrum as input and follows the matter: the Lagrangian map $\mathbf{x} = \mathbf{q} + \mathbf{\Psi}$, the Zel'dovich approximation and its second-order extension, why collapse makes sheets and filaments rather than spheres, and how a threshold applied to a Gaussian field yields linear galaxy bias. The level is advanced undergraduate. Derivations are carried out rather than quoted, and two hands-on sessions and a solutions note accompany the lectures.

Suggested reading. Three textbooks, each better than these notes at something.

1. S. Dodelson & F. Schmidt, *Modern Cosmology*, 2nd ed. (Academic Press, 2020). The closest match to this course, and the one to read if you read only one: Ch. 12 covers Lecture 1 from the same starting point, with its assumptions set out explicitly.
2. D. Baumann, *Cosmology* (Cambridge University Press, 2022). Go here for where the initial conditions come from, namely inflation and the primordial spectrum, which these lectures take as given.
3. D. Huterer, *A Course in Cosmology: From Theory to Practice* (Cambridge University Press, 2023). Go here for the observational side: how the quantities we compute are actually measured, and from what.

How the blocks work. **Boxes** hold material that was not covered at the board, about a fifth of these notes. None of it is needed to follow the main thread, and all of it can be skipped on a first reading. **Derivations** are a few lines of algebra posed at the board and done during the break, not homework, and **Exercises** come at the end of each lecture. Both are solved in the solutions note, distributed as an ancillary file with the arXiv version of these notes.

Contents

Lecture 1: From primordial fluctuations to the linear power spectrum	2
1.1 The Newtonian equations and their assumptions	4
1.2 The pressureless fluid, and how perturbations grow	7
1.3 Gaussian random fields and the power spectrum	7
1.4 The primordial spectrum	9
1.5 Horizon crossing, and the shape of $P_L(k)$	10

Lecture 2: From the power spectrum to the cosmic web	13
2.1 The Lagrangian picture	13
2.2 The Zel'dovich approximation	14
2.3 Why a web, and not a collection of spheres	15
2.4 Shell crossing, and where the approximation fails	16
2.5 Second order: 2LPT and initial conditions	17
2.6 Galaxies: linear bias	18
Appendices	20
A Glossary	20

Lecture 1: From primordial fluctuations to the linear power spectrum

The setting and conventions. Fixed for the whole course: a spatially flat Friedmann–Lemaître–Robertson–Walker (FLRW) universe, which is homogeneous, isotropic and expanding, carrying *scalar* perturbations only. The three perturbation types decouple at linear order: vectors decay, and tensors (gravitational waves) do not source density perturbations, so structure formation is a scalar story. The unperturbed metric is

$$ds^2 = a^2(\tau) [-d\tau^2 + \delta_{ij} dx^i dx^j],$$

written in comoving coordinates $\mathbf{x}$, which ride along with the expansion: a galaxy at rest keeps the same $\mathbf{x}$ while its physical separation from us, $a\mathbf{x}$, grows. Here τ is *conformal* time, defined by $dt = a d\tau$, so that $\tau = \int dt/a$: an integral, not t/a . Overdots mean $\partial/\partial\tau$ throughout, at fixed comoving position for a field and at fixed element for a trajectory, and $\mathcal{H} \equiv d \ln a / d\tau = aH$ is the expansion rate in that time. Spatial indices are raised and lowered with δ_{ij} , so a repeated pair is summed regardless of placement.

Box 1. Conformal time. Conformal time is used because of the form the metric then takes: with $d\tau$ and $d\mathbf{x}$ appearing symmetrically, radial light rays obey $d\mathbf{x} = \pm d\tau$, so light travels at unit comoving speed and an interval of conformal time *is* a comoving distance. That is what will make the horizon a clean notion in Section 1.5.

Structure is what happens when this is not quite true. The metric picks up small perturbations, and the matter density departs from its mean by

$$\delta(\mathbf{x}, \tau) \equiv \frac{\rho(\mathbf{x}, \tau) - \bar{\rho}(\tau)}{\bar{\rho}(\tau)},$$

with $|\delta| \ll 1$ early on and of order unity by the time there are galaxies to count. How δ gets from one to the other is the whole course.

Derivation 1. Show, from the definition above, that the density perturbation $\delta\rho \equiv \rho - \bar{\rho}$ averages to zero at every time, $\overline{\delta\rho}(\tau) = 0$, and hence that $\delta(\mathbf{k}=0) = 0$. Nothing about gravity or expansion is needed. The lesson to bring back is that in the split $\rho = \bar{\rho}(1 + \delta)$ it is $\bar{\rho}$ that carries the mean, and δ carries none of it. The hands-on session sets the $\mathbf{k} = 0$ mode to zero by hand for exactly this reason.

Position space and Fourier space. Both hold the same information. In position space $\delta(\mathbf{x})$ is a lumpy field, where intuition lives, and where Lecture 2's collapsing sheets are easiest to picture. In Fourier space we resolve it into plane waves.

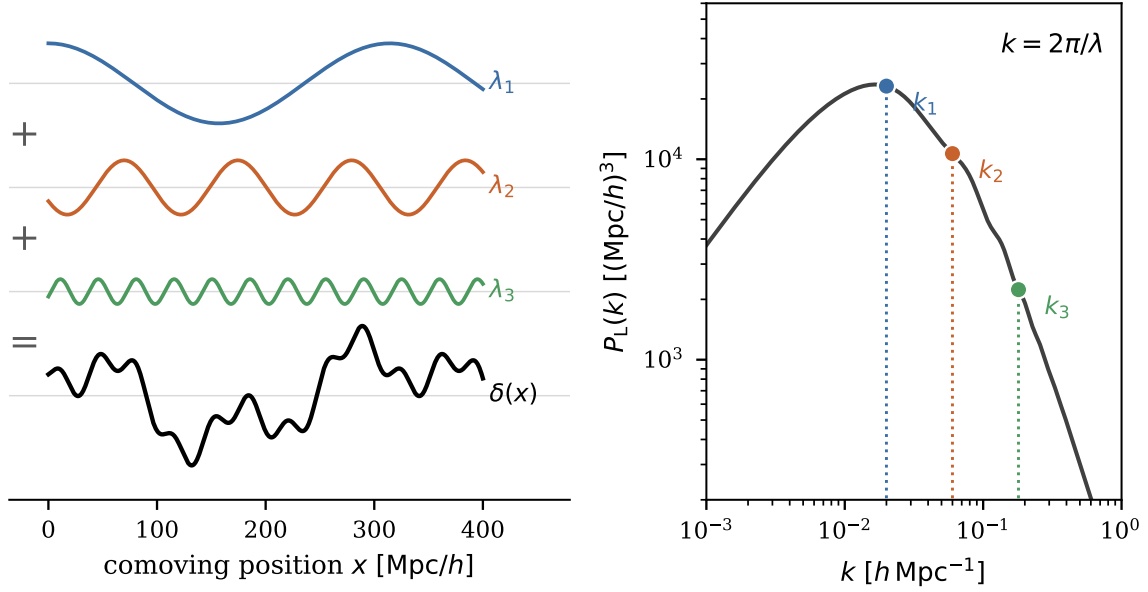


Figure 1: The same field, held two ways. *Left*: three plane waves of decreasing wavelength, $\lambda_1 > \lambda_2 > \lambda_3$, added into a lumpy density field $\delta(x)$. This is a one-dimensional slice for legibility; the real field is three-dimensional and each mode carries a direction $\hat{\mathbf{k}}$ as well as a wavelength. *Right*: the same three modes as wavenumbers $k_i = 2\pi/\lambda_i$, placed on the linear power spectrum: nothing more exotic than amplitude against scale (defined properly in Section 1.3, and computed in Section 1.5). The longest wave has the smallest k . Colours identify the same mode across the two panels.

One warning, because the picture invites the wrong conclusion. The longest wave is drawn with the largest amplitude, correctly: a single mode's amplitude goes as $\sqrt{P_L(k)}$, and P_L falls beyond its peak. But that is *not* the same as large scales dominating the lumpiness. Short-wavelength modes are far more numerous, and the fluctuation power per logarithmic interval, $\Delta^2(k) = k^3 P_L(k)/2\pi^2$, rises steeply across these three modes: 0.009, 0.12, 0.66. Small scales carry more, which is why nonlinearity reaches them first, and why almost everything difficult in this course happens at large k .

Start in a periodic box of side L , which is what the hands-on session uses. The field is a discrete sum, one term per mode,

$$\delta(\mathbf{x}) = \sum_{\mathbf{k}} \delta_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{x}}, \quad \mathbf{k} = \frac{2\pi}{L} \mathbf{n}, \quad \mathbf{n} \in \mathbb{Z}^3,$$

with the spacing $2\pi/L$ set by the box. Let $L \rightarrow \infty$: the spacing closes up, the sum becomes an integral, and we have the convention used throughout,

$$\delta(\mathbf{k}) = \int d^3x e^{-i\mathbf{k}\cdot\mathbf{x}} \delta(\mathbf{x}), \quad \delta(\mathbf{x}) = \int_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{x}} \delta(\mathbf{k}), \quad \int_{\mathbf{k}} \equiv \int \frac{d^3k}{(2\pi)^3}.$$

Each mode is a plane wave of comoving wavelength $\lambda = 2\pi/k$, so large k means *small* scales: $k = 0.1 h \text{ Mpc}^{-1}$ is $\lambda \simeq 63 h^{-1} \text{ Mpc}$, a few cluster diameters.

Why bother: because the background is homogeneous, the linearized equations have coefficients independent of position. A plane wave stays a plane wave, and each $\delta(\mathbf{k})$ evolves independently of every other: one partial differential equation in three dimensions becomes one ordinary differential equation per $\mathbf{k}$. That is what makes linear theory solvable. Figure 1 draws it.

1.1 The Newtonian equations and their assumptions

From here on everything is Newtonian: a fluid in an expanding box, no metric in sight. Three equations do all the work, and they rest on one gauge choice and four numbered assumptions, A1–A3. Keep the list in a corner of the board: the assumptions do different jobs, fail at different times, and exactly one will be dropped in Lecture 2.

The gauge choice. The arena was fixed in the prelude: flat FLRW, scalar perturbations only. What comes before any assumption is a *choice*, and it restricts no physics. Fix the gauge completely by writing the perturbed metric in conformal Newtonian (longitudinal) form [1],

$$ds^2 = a^2(\tau) \left[-(1 + 2\psi) d\tau^2 + (1 - 2\phi) \delta_{ij} dx^i dx^j \right]. \quad (1.1)$$

Here $\psi(\mathbf{x}, \tau)$ perturbs g_{00} and is what a slow particle falls down, while $\phi(\mathbf{x}, \tau)$ perturbs the spatial metric and is what appears in the field equation. Both are fields on space and time, exactly like δ : they vary from place to place and they evolve, and the rest of this lecture computes how.

Box 2. The factors of two in the metric. The factors of 2 and the relative sign are both conventions, chosen to keep every later formula clean. The 2 normalizes ψ : with it, $\Gamma^i_{00} = \partial_i \psi$ exactly at first order, so a slow particle obeys $\ddot{x}^i = -\partial_i \psi$ and ψ is the Newtonian potential rather than twice it (Exercise 1.1). Write $1 + \psi$ instead and you get $\Gamma^i_{00} = \frac{1}{2} \partial_i \psi$, so a factor of one half propagates into Newton’s law and into the Poisson equation and never leaves. The relative minus sign does the analogous job further down: it is what makes vanishing anisotropic stress read $\phi = \psi$, rather than $\phi = -\psi$, so that the two potentials can be replaced by the single Φ .

A1: weak fields, $|\phi|, |\psi| \ll 1$. Work to first order in both potentials. An observation as much as an assumption: the potentials are of order 10^{-5} on every scale we will meet, at every time. Note what A1 does *not* mention: the density contrast δ .

A2: cold matter. The matter is cold dark matter and baryons, pressureless and slow, $u \ll 1$. Newtonian gravity has one potential, not two, and A2 is what closes the gap: the traceless part of Einstein’s equations sources $\phi - \psi$ through anisotropic stress alone, which cold matter does not carry, so for us the two potentials coincide, $\phi = \psi \equiv \Phi$.

Box 3. Anisotropic stress, and neutrinos. What must vanish is the *total* anisotropic stress [1, Eq. (23d)], and free-streaming neutrinos spoil it at the 15% level during radiation domination. That contribution fades as matter comes to dominate, so in the matter era $\phi = \psi$ is accurate.

A3: sub-horizon modes. A4: linear contrasts. The last two arrive with the system itself: keep only $k \gg \mathcal{H}$ (A3), and linearize in the density contrast, $|\delta| \ll 1$ (A4). A3 is permanent. A4 is temporary, and is the one assumption Lecture 2 drops, for a reason given at the end of this section.

A3 deserves a word, since we lean on it constantly. Because $\mathcal{H}$ is a rate per unit conformal time, $1/\mathcal{H}$ is a *time*, roughly how long the universe takes to expand appreciably; and because light travels at unit comoving speed in conformal time, the same number is also a *length*, the comoving distance light covers in that time. That length is the *comoving Hubble radius*, loosely “the horizon”. A mode is *sub-horizon* when $k \gg \mathcal{H}$, meaning shorter than the horizon, so that signals cross it many times while the universe expands once and gravity acts coherently across it. It is *super-horizon* when $k \ll \mathcal{H}$, where nothing crosses it at all. Section 1.5 is about what happens to a mode when it changes from one to the other.

With the list complete: we are describing a **self-gravitating fluid**, matter moving in a potential that it sources itself. That fixes what the system must contain. A fluid conserves mass and

momentum, which is two equations; and because the gravity is its own, a third is needed to tie the potential back to the density. Those three, one at a time.

Mass conservation. The contrast changes only because matter flows. For the peculiar velocity $\mathbf{u} \equiv d\mathbf{x}/d\tau$, mass conservation reads

$$\dot{\delta} + \nabla \cdot [(1 + \delta) \mathbf{u}] = 0, \quad (1.2)$$

in which dilution by the expansion never appears; it is already inside $\bar{\rho}$, which is the content of Derivation 1. At linear order (A4) the product term drops and (1.2) becomes $\dot{\delta} = -\theta$, with $\theta \equiv \nabla \cdot \mathbf{u}$. (The relativistic version carries one more term, a dilation $3\dot{\phi}$; A3 suppresses it, as Exercise 1.3 shows.)

Newton's second law. For a pressureless fluid (A2) there is no separate Euler equation to import: Euler *is* Newton's second law applied to fluid elements. Follow one element, so that $\mathbf{x}(\tau)$ is its trajectory. To first order in Φ (A1),

$$\ddot{\mathbf{x}} + \mathcal{H} \dot{\mathbf{x}} = -\nabla \Phi. \quad (1.3)$$

Since $\mathbf{u} = \dot{\mathbf{x}}$, this *is* the fluid's momentum equation, $\dot{\mathbf{u}} + \mathcal{H}\mathbf{u} = -\nabla \Phi$, written per element rather than per point. The density cannot appear in it, because momentum conservation is a statement about motion. Eliminate $\mathbf{u}$ and Φ between the three members and you get a single equation for δ alone; that is (1.6), and Section 1.2 does it.

The new term is the drag $\mathcal{H}\dot{\mathbf{x}}$, and it is not friction. Set $\Phi = 0$ and (1.3) says $\dot{\mathbf{x}} \propto 1/a$: free particles coast to rest in comoving coordinates as their peculiar velocities redshift away. Two exercises reach it from opposite directions: Exercise 1.1 comes down from the geodesic equation of (1.1), and Exercise 1.2 builds up from Newtonian momentum conservation without relativity at all. Lecture 2 is built on this equation: it follows elements rather than fields, and that change of variable is the whole of the Lagrangian picture. For now we need only its divergence: with $\mathbf{u} = \dot{\mathbf{x}}$ and $\theta = \nabla \cdot \mathbf{u}$, at linear order (A4) $\dot{\theta} = -\mathcal{H}\theta - \nabla^2 \Phi$.

The Poisson equation. What closes the system is the field equation, and here A3 does its work: every term carrying a factor of $\mathcal{H}$ where its neighbours carry k is suppressed by $(\mathcal{H}/k)^2$, and the energy constraint of general relativity collapses to the Poisson equation, $\nabla^2 \Phi = 4\pi G a^2 \bar{\rho} \delta = \frac{3}{2} \Omega_m(\tau) \mathcal{H}^2 \delta$, the last equality by the Friedmann equation. Exercise 1.3 carries out the reduction and puts a number on what was dropped.

Assembled, at linear order (A4):

$$\dot{\delta} = -\theta, \quad \dot{\theta} = -\mathcal{H}\theta - \nabla^2 \Phi, \quad \nabla^2 \Phi = \frac{3}{2} \Omega_m(\tau) \mathcal{H}^2 \delta. \quad (1.4)$$

Ma & Bertschinger [1, §5] give the unapproximated system; Dodelson & Schmidt [9, Ch. 12] set out this same starting point with its assumptions listed.

Derivation 2. Assemble (1.4) for yourself: linearize (1.2) to get the first member, take the divergence of (1.3) to get the second, and write down the third. Keep track of which assumption each step spends, and check yourself against the discussion that follows.

Box 4. Energy conservation for cold matter. Energy conservation is not a separate equation here, for three reasons that are the same reason seen from three sides.

For cold matter, mass conservation *is* energy conservation. The energy density is rest mass; the kinetic part is smaller by $u^2 \ll 1$ (A2). So (1.2) already did that job.

There is also no independent energy equation available to write. A fluid with pressure needs one, since something must say how the pressure evolves, and that is what closes the system. Coldness closes it instead: with no velocity dispersion the chain of moment equations truncates after Euler, which is why two fluid equations suffice here where a gas would need three. This is also the assumption that expires. Once orbits cross, one velocity per point is no longer true and the truncation fails, which is where Lecture 2 stops.

And least obviously, you did use it. The relativistic statement that energy density curves spacetime is the time–time Einstein equation, called the *energy constraint*, and its sub-horizon limit is the third member of (1.4). Poisson is the energy equation, in the only form general relativity offers; Exercise 1.3 works with it unapproximated.

Which assumption lives where. The three members of (1.4) do not lean on the same legs. *Continuity* is the most robust: mass conservation, needing only A4 to linearize and one use of A3, to drop the relativistic dilation term $3\dot{\phi}$. *Euler*, in the form $\dot{\theta} = -\mathcal{H}\theta - \nabla^2\Phi$, needs A2 and A1, cold and slow matter in a weak field, but *no* use of A3 at all: for cold matter it is exact at linear order in this gauge. *Poisson* is where A3 does its real work: the relativistic energy constraint carries extra terms of relative size $(\mathcal{H}/k)^2$, and dropping them is what turns a constraint of general relativity into Newton’s field equation. Exercise 1.3 makes this quantitative. And the $\phi = \psi$ of A2 underwrites the system as a whole rather than any single member: Euler feels the time–time potential while Poisson sources the spatial one.

The smallness of Φ . A3 has a second consequence, and the whole of Lecture 2 rests on it.

It keeps the metric linear. Inverting the Poisson equation,

$$|\Phi| \sim \frac{3}{2} \Omega_m \left(\frac{\mathcal{H}}{k} \right)^2 \delta, \quad (1.5)$$

the potential is the density contrast suppressed by two powers of $\mathcal{H}/k$. At $k = 0.1 \, h \, \text{Mpc}^{-1}$ that factor is about 10^{-5} , so even a patch with $\delta \sim 100$ carries $|\Phi| \sim 5 \times 10^{-4}$. Gravity stays weak precisely where clustering becomes strong. This is what licenses everything that follows. We will let δ grow past unity and resum it to all orders through displacements, while never going beyond linear order in the metric. Those two statements sound contradictory and are not: Φ and δ are separated by $(\mathcal{H}/k)^2$, and it is δ alone that becomes large. Dropping A4 leaves A1 intact.

Derivation 3. Derive (1.5) yourself: write the third member of (1.4) in Fourier space and show that $|\Phi| = \frac{3}{2} \Omega_m(\tau) (\mathcal{H}/k)^2 |\delta|$, so that sub-horizon (A3) the weak-field condition (A1) on Φ is guaranteed rather than assumed. Then put in numbers ($\mathcal{H}_0 \simeq 3.3 \times 10^{-4} \, h \, \text{Mpc}^{-1}$, $k = 0.1 \, h \, \text{Mpc}^{-1}$, $\delta = 100$) and check the 5×10^{-4} quoted above.

Box 5. Why δ needs a gauge. The potential Φ is unambiguous, but δ is not, and fixing the gauge is what makes it meaningful. Writing $\delta = (\rho - \bar{\rho})/\bar{\rho}$ requires $\bar{\rho}$ at the same time as ρ , and in general relativity “the same time” is a choice of how to slice spacetime into spatial surfaces, not a fact about the universe. Shift that slicing infinitesimally, $\tau \rightarrow \tau + \xi^0(\mathbf{x}, \tau)$. Since $\bar{\rho}$ depends on time it moves too, by $\dot{\bar{\rho}}\xi^0$, and the background continuity equation $\dot{\bar{\rho}} = -3\mathcal{H}(1+w)\bar{\rho}$ then gives

$$\delta \longrightarrow \delta + 3\mathcal{H}(1+w)\xi^0.$$

That is the whole statement: δ changes when you re-slice, so it is not by itself a physical quantity. It is also why the same mode grows as $(k\tau)^2$ in synchronous gauge and sits frozen in the Newtonian gauge until horizon entry [1, §7], and why “do perturbations grow outside the horizon?” has no gauge-independent answer.

There is a repair, and it explains A3 from a second direction. The combination $\Delta \equiv \delta + 3\mathcal{H}(1+w)v$, with v the velocity potential ($\mathbf{u} = \nabla v$), is gauge-invariant, since the shift above cancels against the one v picks up, and it obeys $-k^2\Phi = \frac{3}{2}\Omega_m\mathcal{H}^2\Delta$ exactly, on all scales, with no sub-horizon limit anywhere. Our Poisson equation is that one with Δ replaced by δ . The error is the difference: $\theta \sim \mathcal{H}\delta$ gives $v \sim \mathcal{H}\delta/k^2$, so $\Delta - \delta \sim (\mathcal{H}/k)^2\delta$. Sub-horizon the two agree and δ is as good as gauge-invariant; super-horizon the correction is the whole answer.

Φ carries no such disease: (1.1) fixes the slicing completely, leaving no residual freedom, so Φ is physical on every scale: it is the Bardeen potential under another name. That is exactly why Section 1.5 follows Φ and not δ when it has to cross the horizon.

Where relativity actually enters. Everything these lectures *compute* is Newtonian and sub-horizon: the system (1.4), the linear growth it implies, and the whole of Lecture 2. Beyond the FLRW background itself, general relativity appears exactly three times, each as an input rather than a calculation: the metric (1.1) with $\phi = \psi$, which defines Φ ; the primordial spectrum of Section 1.4, which is inflation's output; and the one super-horizon fact that Φ holds still outside the horizon (Section 1.5), used only to carry the initial condition intact to horizon entry. One dynamical ingredient is quoted rather than derived: the acoustic oscillation of the radiation fluid, which needs pressure and so lies outside (1.4). Everything else is the three equations above.

1.2 The pressureless fluid, and how perturbations grow

Everything needed is already in (1.4). One check first: the curl of (1.3) at linear order gives $\dot{\mathbf{w}} + \mathcal{H}\mathbf{w} = 0$ for the vorticity $\mathbf{w} = \nabla \times \mathbf{u}$, so rotation decays as $1/a$ and stays zero if it starts zero, so the velocity field is fully specified by its divergence θ . Now combine all three: differentiate $\dot{\delta} = -\theta$, substitute $\dot{\theta}$, and eliminate $\nabla^2 \Phi$ with Poisson. No coefficient in the result carries a k , so the position dependence factors out and every mode grows by one common factor,

$$\ddot{D} + \mathcal{H}\dot{D} - \frac{3}{2}\Omega_m(\tau)\mathcal{H}^2 D = 0. \quad (1.6)$$

The growing solution $D_+(\tau)$ defines linear growth, $\delta^{(1)}(\mathbf{x}, \tau) = D_+(\tau)\delta_0(\mathbf{x})$; in Einstein–de Sitter (EdS, $\Omega_m = 1$) $D_+ = a$ and the decaying mode is $a^{-3/2}$. The velocity follows from continuity, and introduces the growth rate f ,

$$\theta^{(1)} = -\mathcal{H}f\delta^{(1)}, \quad f \equiv \frac{d \ln D_+}{d \ln a} \simeq \Omega_m^{5/9}(\tau) \quad (\text{flat } \Lambda\text{CDM}), \quad (1.7)$$

with $f = 1$ in EdS. The often-quoted exponent 0.55 [11] agrees with 5/9 at the sub-percent level over viable cosmologies. See [10] for a review of how the growth of structure is measured.

Exercise 1.4 gets both solutions by hand in Einstein–de Sitter, which is the case worth doing, since everything else is a numerical perturbation on it.

Two modes, so the general solution mixes them: $\delta = A(\mathbf{x})D_+(\tau) + B(\mathbf{x})D_-(\tau)$, with A and B fixed by the initial density *and* the initial velocity, since a second-order equation needs two pieces of initial data. Relative to the growing mode the second term falls as $D_-/D_+ \propto a^{-5/2}$, so after enough expansion only $A(\mathbf{x})D_+(\tau)$ survives. That is what licenses writing $\delta^{(1)} = D_+(\tau)\delta_0(\mathbf{x})$ above: one function of position times one of time, rather than two.

What it does not erase is the initial conditions. $A(\mathbf{x})$ is the primordial field, remembered exactly, and computing it is what Lecture 1 does. What is forgotten is only the other half of the initial data: whatever part of the start did not lie on the growing solution. This is why initial conditions for a simulation can be laid down using the growing mode alone (Section 2.5), and why a slightly wrong start relaxes onto the right answer instead of propagating.

The growing mode is remarkable for how *slow* it is. In a static medium an overdensity grows exponentially; expansion converts that runaway into a power law. The Friedmann equation makes $\mathcal{H}^2 \sim Ga^2\bar{\rho}$, so the drag $\mathcal{H}\dot{D}$ and the source $\frac{3}{2}\Omega_m\mathcal{H}^2 D$ in (1.6) are always comparable: gravity wins, but only just. That is why perturbations of order 10^{-3} at recombination are of order unity now, after a factor of a thousand in a , instead of having collapsed long ago, and why perturbation theory is useful across five decades of expansion.

1.3 Gaussian random fields and the power spectrum

Inflation delivers, to very good approximation, Gaussian initial conditions: the linear field $\delta^{(1)}(\mathbf{k})$ is a Gaussian random field, meaning that any finite set of its Fourier modes is jointly Gaussian with zero mean. Such a field is characterized completely by its power spectrum,

$$\langle \delta^{(1)}(\mathbf{k}) \delta^{(1)}(\mathbf{k}') \rangle = (2\pi)^3 \delta_D(\mathbf{k} + \mathbf{k}') P_L(k), \quad (1.8)$$

where two distinct assumptions are at work. Statistical *homogeneity*, invariance under translations, is what forces the delta function, so that modes whose wavevectors do not sum to zero are uncorrelated. Statistical *isotropy* is a separate statement, invariance under rotations, and it is what makes P_L a function of the magnitude $k = |\mathbf{k}|$ alone rather than of the vector $\mathbf{k}$. Note what is being predicted. Two seeds give completely different universes, because the phases are random and carry no cosmology. P_L is the *variance*, and the variance is the prediction. We recover it from our single universe by averaging $|\delta(\mathbf{k})|^2$ over the modes of equal $|\mathbf{k}|$, which are independent draws from one distribution. The largest scales have the fewest such modes and are therefore the worst measured: cosmic variance is the price of owning one universe.

Box 6. The two-point correlation function. In position space the same information sits in the two-point correlation function, the excess probability over random of finding matter at two points separated by $\mathbf{r}$,

$$\xi(r) \equiv \langle \delta(\mathbf{x}) \delta(\mathbf{x} + \mathbf{r}) \rangle = \int_{\mathbf{k}} P_L(k) e^{i\mathbf{k} \cdot \mathbf{r}} = \frac{1}{2\pi^2} \int_0^\infty dk k^2 P_L(k) \frac{\sin kr}{kr},$$

the last step doing the angular integral for an isotropic spectrum. So ξ and P_L are a Fourier pair and carry identical information, the same statement as the prelude's, one level up: a field can be held in position space or in Fourier space, and so can its statistics.

Two things follow. At zero separation $\xi(0) = \langle \delta^2 \rangle = \int_{\mathbf{k}} P_L(k)$, the variance of the field, but treat that with care. Because the power per logarithmic interval $k^3 P_L/2\pi^2$ *rises* with k , the integral is dominated by the smallest scales and keeps growing as you push the upper limit out; an unqualified variance is not a number. This is precisely why amplitudes are quoted smoothed, which is what the window in (1.9) is for. On a grid the upper limit is the Nyquist frequency and the integral is finite, which is what lets the hands-on session check its normalization against it. And the baryon wiggles of Section 1.5 become a single bump: an oscillation of period $2\pi/r_d$ in P_L transforms into one feature in ξ at $r = r_d \simeq 100 h^{-1} \text{Mpc}$. Surveys quote whichever is more convenient; it is the same physics either way.

Box 7. Generating a Gaussian field. One pair of modes is exempt from the statement that distinct modes are uncorrelated: $\mathbf{k}' = -\mathbf{k}$, where the delta function fires. Since the field is real, $\delta(-\mathbf{k}) = \delta(\mathbf{k})^*$, so that case is $\langle |\delta(\mathbf{k})|^2 \rangle$: a variance, and necessarily non-negative. It is not a counterexample to “uncorrelated” but the definition of P_L itself.

Generating a realization. That a Gaussian field is fixed completely by P_L is a practical statement as much as a formal one, and it is what the hands-on session begins with. On a periodic grid the recipe is three lines:

1. for each mode $\mathbf{k}$, draw a complex amplitude with zero mean and variance $P_L(k)/V$, where V is the box volume — equivalently, draw the modulus from a Rayleigh distribution and the phase uniformly on $[0, 2\pi)$. That volume factor is the step that most often goes wrong, and it fails silently: the field looks right and every number computed from it is off;
2. impose reality, $\delta(-\mathbf{k}) = \delta(\mathbf{k})^*$, so that the field you get back is real rather than complex;
3. inverse transform.

Nothing else is needed. Two realizations drawn with different random seeds are different universes with identical statistics, which is why a fixed seed is the right way to compare cosmologies: change n_s with the seed held, and every difference you see is physics rather than luck.

Normalization: σ_8 . The spectrum has a shape and an overall size, and the size is conventionally quoted as σ_8 : the rms fluctuation of the linear density field smoothed on spheres of radius $8 h^{-1} \text{Mpc}$,

$$\sigma_8^2 = \int \frac{d^3 k}{(2\pi)^3} P_L(k) |W(kR)|^2, \quad R = 8 h^{-1} \text{Mpc}, \quad (1.9)$$

with W the Fourier transform of a top-hat sphere. The radius is historical rather than fundamental, since it is roughly where $\sigma \simeq 1$ and so marks the scale that is going nonlinear now, but the definition is what normalizes every spectrum you will plot. Measured, $\sigma_8 \simeq 0.81$.

1.4 The primordial spectrum

Everything so far has been about how a perturbation *evolves*. Where the perturbations came from is the business of the lectures you have just had, and we need only their output: a nearly *scale-invariant* spectrum of curvature perturbations $\mathcal{R}$, quoted as usual in dimensionless form,

$$\Delta_{\mathcal{R}}^2(k) \equiv \frac{k^3 P_{\mathcal{R}}(k)}{2\pi^2} = A_s \left(\frac{k}{k_0} \right)^{n_s-1}, \quad n_s \simeq 0.965, \quad (1.10)$$

with A_s an amplitude and k_0 an arbitrary pivot scale. The exponent is written $n_s - 1$ so that $n_s = 1$ is exactly the scale-invariant case. This is Dodelson & Schmidt's Eq. (7.99); much of the inflation literature writes ζ for the same quantity.

Box 8. The pivot scale k_0 . The pivot is a bookkeeping choice, not physics. A tilted power law has no single amplitude, since $\Delta_{\mathcal{R}}^2$ takes a different value at every k , so A_s has to be quoted *somewhere*, and k_0 is that somewhere: A_s is by definition $\Delta_{\mathcal{R}}^2(k_0)$. Move the pivot and A_s moves with it, by $(k'_0/k_0)^{n_s-1}$, leaving $\Delta_{\mathcal{R}}^2(k)$ unchanged at every k . Planck's $A_s = 2.1 \times 10^{-9}$ at $k_0 = 0.05 \text{ Mpc}^{-1}$ would be 2.35×10^{-9} had they chosen 0.002 Mpc^{-1} ; the spectrum is the same spectrum. The value is picked so that the data constrain A_s and n_s close to independently, which puts it near the middle of the measured range of scales.

Two practical warnings. That 0.05 is in Mpc^{-1} , not $h \text{ Mpc}^{-1}$: in the units these lectures use it is $0.074 h \text{ Mpc}^{-1}$, and reading it as $0.05 h \text{ Mpc}^{-1}$ misplaces the pivot by a factor $1/h$. And these lectures never need its value at all, because we fix A in (1.12) with σ_8 rather than with A_s , since the pivot cancels out of everything downstream.

Box 9. The curvature perturbation $\mathcal{R}$. $\mathcal{R}$ is a perturbation to the spatial curvature of a constant-time slice, measured on slices where the matter is at rest: the spatial metric reads $a^2(1 + 2\mathcal{R})\delta_{ij}$, so $\mathcal{R}$ is a local, dimensionless shift in how much the universe has expanded at that point. Its virtue is that it does not evolve outside the horizon, in any background, so inflation's output is quoted in it: it can be handed forward unchanged, across the radiation era and into the matter era, to the moment the mode enters and physics resumes.

Turning it into δ is two steps. First, deep in matter domination the potential is a fixed fraction of it, $|\Phi| = \frac{3}{5}|\mathcal{R}|$, a pure number with no k in it. Second, Poisson converts the potential into a density, and *that* is where the two powers of k come from. Together, for modes well inside the horizon,

$$\delta_m(k, a) = \frac{2}{5} \frac{k^2}{\Omega_m H_0^2} \mathcal{R}(k) T(k) D_+(a),$$

which is Dodelson & Schmidt's Eq. (8.7): the $2/5$ is $\frac{2}{3} \times \frac{3}{5}$, the k^2 is Poisson, T handles modes that entered before equality, and D_+ carries the growth. Square it and the k^4 against $P_{\mathcal{R}} \propto k^{n_s-4}$ gives $P_L \propto k^{n_s} T^2 D_+^2$ — (1.12), with $A = 8\pi^2 A_s k_0^{1-n_s} / 25 H_0^4 \Omega_m^2$.

One warning when you open the book: they write the spatial metric as $a^2(1 + 2\Phi)$ where (1.1) writes $a^2(1 - 2\phi)$, so their Φ is minus ours. The relation above is sign-consistent as written; the intermediate potentials are not.

Box 10. Why n_s is not exactly one. Scale invariance means equal power in every logarithmic interval of k , which is what $n_s = 1$ would give. Inflation cannot deliver it exactly: inflation has to end, so the expansion rate drifts while modes are leaving the horizon, and the spectrum tilts. The measured few-percent deficit is a genuine prediction rather than a fitted convenience, and it is one of the reasons to believe the picture.

Two consequences carry forward. The primordial field is Gaussian to excellent approximation, which is what Section 1.3 assumed. And (1.10) becomes a *matter* spectrum by two steps that all but cancel. Reading $P_{\mathcal{R}}$ off the definition divides by k^3 ; the Poisson equation relating curvature to density then multiplies by k^4 , since it costs two powers of k in the field and so four in the spectrum. Net one power of k :

$$P_{\text{prim}}(k) = A k^{n_s}, \quad (1.11)$$

a featureless power law with no scale anywhere in it. Note where scale invariance actually sits: $n_s = 1$ means $P_{\text{prim}} \propto k$, not P_{prim} constant.

1.5 Horizon crossing, and the shape of $P_L(k)$

Write the linear spectrum as the primordial one times a *transfer function* that carries all the processing:

$$P_L(k, z) = A k^{n_s} T^2(k) D_+^2(z) \quad (1.12)$$

with $T \rightarrow 1$ on the largest scales by convention and A fixed by one measured amplitude, for which we use σ_8 . The three factors are three separate pieces of physics: inflation sets k^{n_s} , the growth factor D_+^2 carries the time dependence of Section 1.2, and $T^2(k)$ carries everything in between. This is the formula you will code in the hands-on session, and the question now is what shapes T .

One thing does: **when a mode entered the horizon.**

The horizon, and what crosses it. Section 1.1 defined the comoving Hubble radius $1/\mathcal{H}$ and used it to separate sub-horizon modes from super-horizon ones. The point now is that the separation is not permanent: $1/\mathcal{H}$ changes with time while a comoving k does not, so a given mode can start on one side and end on the other. It does, and that is the mechanism.

Box 11. H versus $\mathcal{H}$. Overdots here are $\partial/\partial\tau$ throughout, so $\dot{a}/a$ is $\mathcal{H}$ and not the Hubble parameter $H = d \ln a / dt$, and it is also why the growth equation (1.6) carries $\mathcal{H}\dot{D}$ where its cosmic-time form carries $2H dD/dt$.

You met this quantity in the previous lectures, running the other way. During inflation $1/\mathcal{H}$ *shrinks*, which is how modes exit and how their fluctuations are frozen in as the initial conditions of Section 1.4. Afterwards it grows, through radiation and matter domination alike, so a mode, labelled by a comoving k that never changes, starts super-horizon and *enters* when $k = \mathcal{H}$. Nothing about the mode changes at that moment; the horizon moves past it.

What follows is what that crossing does, and Fig. 2 is the picture to keep in mind while it happens.

Follow the potential Φ rather than the density. It is unambiguous, since Section 1.1 fixed the gauge completely, and it obeys three rules.

Outside the horizon, Φ holds still. No causal process operates across the mode, so nothing can reshape it. This does not follow from (1.4) — those equations assumed $k \gg \mathcal{H}$ — and it is not used to *evolve* anything: it says there is nothing to evolve, so the mode reaches entry still carrying the primordial amplitude of Section 1.4. Super-horizon is where the initial condition sits, not a phase we integrate through.

Inside the horizon during radiation domination, Φ decays. Sub-horizon the energy constraint is again a Poisson equation, $-k^2\Phi = 4\pi G a^2 \bar{\rho} \delta$. The constraint cares only about what dominates the density, here radiation. The *dynamics* of (1.4) do not carry over: radiation has pressure, which is exactly what “pressureless” excluded, so how δ_r behaves is imported here rather than derived. Two things then happen at once. The prefactor falls: $\bar{\rho}_r \propto a^{-4}$, so $a^2 \bar{\rho}_r \propto a^{-2}$. And the source refuses to grow: radiation pressure resists compression, so δ_r does not collapse but rings, as a sound wave in the tightly coupled photon–baryon plasma. Hence $\Phi \propto a^{-2} \times (\text{oscillation})$. Keep both factors in view. They become the two features of the spectrum: the a^{-2} envelope sets the broadband shape, and the oscillation is the BAO.

Inside the horizon during matter domination, Φ holds still again. Now $a^2 \bar{\rho}_m \propto a^{-1}$ while δ grows as $D_+ \propto a$, and the two cancel exactly. This is why the potential is constant through the whole matter era, and why Φ stayed small in (1.5).

Assembling these. A mode with $k < k_{\text{eq}}$ enters *after* equality, so it never sees the decaying phase and arrives with its primordial amplitude intact, $T \simeq 1$. It does still enter: $k = 10^{-3} h \text{ Mpc}^{-1}$ crosses at $a \simeq 0.03$ and has $T = 0.99$. What matters is that Φ has stopped decaying by then, so entry costs it nothing. Suppression requires entering *while radiation dominates*, not entering.

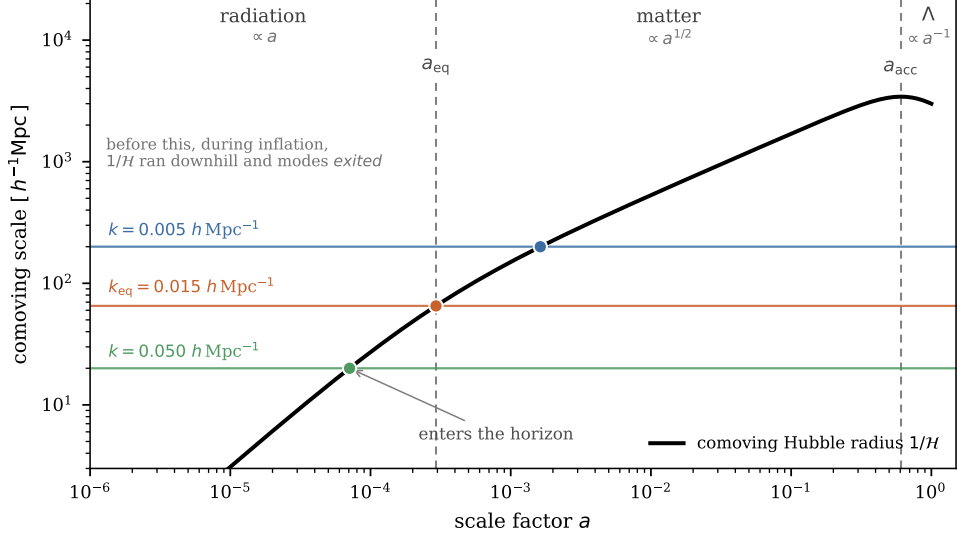


Figure 2: Horizon entry. The heavy curve is the comoving Hubble radius $1/\mathcal{H}$; the coloured lines are three modes, each at its fixed comoving $1/k$. A mode is a horizontal line because $\mathbf{k}$ is a comoving label that never changes. What moves is the horizon, and a mode *enters* where the two meet. The mode entering exactly at matter–radiation equality defines k_{eq} , and the figure is drawn with it computed rather than assumed: $\mathcal{H}(a_{\text{eq}}) = 0.0153 \, h \, \text{Mpc}^{-1}$ for $\Omega_m = 0.31$, $h = 0.676$ and three massless neutrino species, giving $a_{\text{eq}} = 2.9 \times 10^{-4}$ and $z_{\text{eq}} = 3410$.

The curve bends because the figure spans three eras, and $1/\mathcal{H}$ is a different power of a in each: $1/\mathcal{H} \propto a$ under radiation, $a^{1/2}$ under matter, a^{-1} under Λ — log-slopes $+1$, $+\frac{1}{2}$, -1 , which is what the plotted curve measures ($+0.998$ at $a = 10^{-6}$, $+0.514$ at $a = 10^{-2}$, -0.534 today and still steepening). The break at a_{eq} is what makes k_{eq} special, and the slope of $+1$ to its left is exactly the statement $a_{\text{enter}} \propto 1/k$ that (1.13) rests on. The turnaround sits at $a = 0.61$ ($z = 0.64$), which is $(\Omega_m/2\Omega_\Lambda)^{1/3}$, the onset of acceleration. Past it modes leave the horizon again, the same thing inflation did, in the same direction.

A mode with $k > k_{\text{eq}}$ enters at a_{enter} during radiation domination and decays until equality. Nothing happens before a_{enter} or after a_{eq} , and $\Phi \propto a^{-2}$ in between, so the accumulated suppression is that power law read across the window,

$$\frac{\Phi(a_{\text{eq}})}{\Phi_{\text{prim}}} = \left(\frac{a_{\text{enter}}}{a_{\text{eq}}} \right)^2,$$

frozen in from then on: a^{-2} evaluated at the ends of the interval, so the earlier a mode enters the less survives. For the k -dependence, entry means $k = \mathcal{H}(a_{\text{enter}})$ with $\mathcal{H} \propto 1/a$ in radiation domination, giving $a_{\text{enter}} \propto 1/k$. Normalizing on the mode entering at equality, $a_{\text{enter}}/a_{\text{eq}} = k_{\text{eq}}/k$, so

$$T(k) \sim \left(\frac{a_{\text{enter}}}{a_{\text{eq}}} \right)^2 \propto \left(\frac{k_{\text{eq}}}{k} \right)^2. \quad (1.13)$$

That is the shape, from horizon crossing and the Poisson equation alone: flat below the break, falling as k^{-2} above it, with the break at $k_{\text{eq}} \equiv \mathcal{H}(a_{\text{eq}}) \simeq 0.015 \, h \, \text{Mpc}^{-1}$. It is not a fitted parameter — it depends on $\Omega_m h^2$ and the radiation density — and it is the one scale the early universe stamps on the spectrum.

Box 12. Two refinements: Mészáros, and the acoustic wiggles. The cold dark matter is not quite frozen while the potential decays. It grows logarithmically, the *Mészáros effect*, which softens the falloff to $T \propto k^{-2} \ln k$. You can see this in a real transfer function: fitting the Eisenstein & Hu form over $0.5 < k < 5 \, h \, \text{Mpc}^{-1}$ gives a slope of -1.67 , not -2 , and the difference is the logarithm. For the spectrum

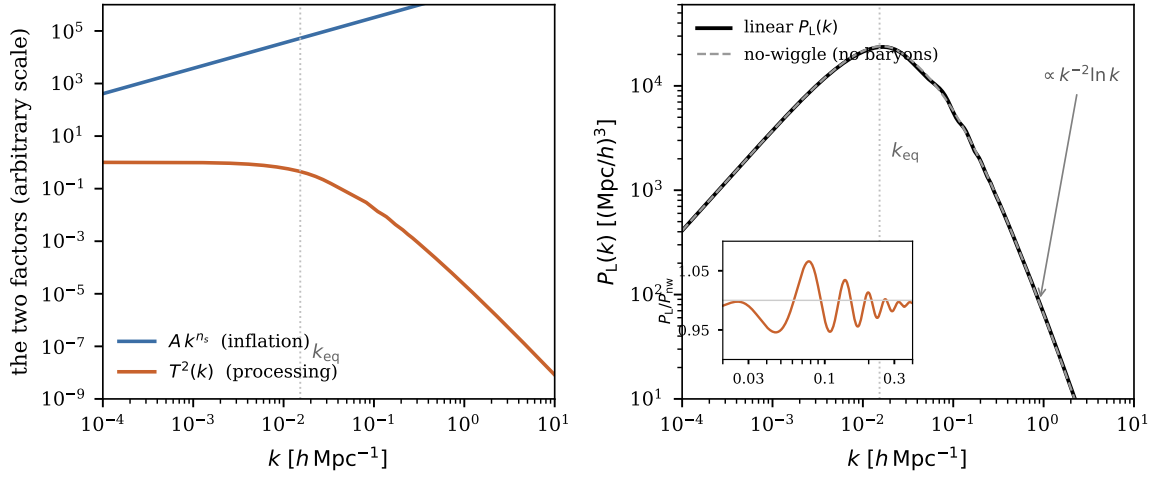


Figure 3: The two factors of $P_L = A k^{n_s} T^2(k)$, and their product. *Left:* inflation supplies a featureless power law with no scale in it; the transfer function supplies everything else: flat below k_{eq} , falling as k^{-2} above it. *Right:* the product, which is the input to everything that follows. The turnover sits at k_{eq} , marked by the dotted line; the high- k slope is k^{-2} softened by the Mészáros logarithm; and the inset shows the baryon wiggles as a ratio to the no-wiggle spectrum, since at a few percent they are invisible on the main panel. Computed with the Eisenstein & Hu transfer function [4], the one you will code in the hands-on session.

itself, (1.12) therefore runs from $P_L \propto k^{n_s}$ below the turnover to $P_L \propto k^{n_s-4} \ln^2 k$ far above it: a measured slope of -2.37 over that same range, still well short of the asymptotic $n_s - 4 = -3.04$, because the logarithm is in no hurry. And notice where that high- k shape comes from: k^{n_s-4} is the shape of $P_{\mathcal{R}}$ itself. The k^2 that Poisson puts into δ is exactly the k^{-2} that the decaying potential takes out, so a small-scale mode ends up with a late-time δ proportional to its primordial $\mathcal{R}$ and no extra scale dependence at all. Exercise 1.5 works it out. And that oscillation survives as the baryon acoustic wiggles: the plasma rings until recombination frees the photons, freezing the wave at a single scale, the sound horizon $r_d \simeq 100 h^{-1} \text{Mpc}$. Baryons are only about a sixth of the matter, so the wiggles ride on the broadband shape at the few-percent level rather than dominating it.

Exercise 1.1. From above. Derive (1.3): write the geodesic equation for a nonrelativistic particle in the metric (1.1), keep terms to first order in the potentials and to leading order in $u \ll 1$, and use $\phi = \psi = \Phi$. Careful: conformal time is not an affine parameter.

Exercise 1.2. From below. Now reach (1.3) with no relativity at all. Start from momentum conservation in physical coordinates, $d^2 \mathbf{r} / dt^2 = -\nabla_r \Phi_{\text{tot}}$ with $\mathbf{r} = a \mathbf{x}$, and split Φ_{tot} into the part sourced by $\bar{\rho}$ and the part sourced by $\bar{\rho} \delta$. (i) Using the Friedmann acceleration equation $\ddot{a} = -\frac{4\pi}{3} G \bar{\rho} a$, show that the background gravity cancels exactly, so that only the peculiar potential survives. Why is that cancellation not an accident of $p = 0$? (ii) Convert to conformal time and recover (1.3). (iii) Where did the drag term come from, and is it a force?

Exercise 1.3. The other limit. Exercise 1.1 took $v \ll 1$ and produced the force law; this one takes $k \gg \mathcal{H}$ and produces the field equation. The linearized Einstein equation that becomes Poisson is the energy constraint [1, Eq. (23a)],

$$\underbrace{k^2 \phi}_{\text{survives}} + \underbrace{3\mathcal{H}(\dot{\phi} + \mathcal{H}\psi)}_{\text{dropped}} = -4\pi G a^2 \bar{\rho} \delta.$$

(i) Reducing it to the Poisson equation of (1.4) needs slightly more than $k \gg \mathcal{H}$: it needs $\dot{\phi}$ not to be anomalously large. Assuming $\dot{\phi} \lesssim \mathcal{H}\phi$, which holds for the growing mode, show that the

discarded group is smaller than the retained one by $\mathcal{O}((\mathcal{H}/k)^2)$. (ii) Put a number on it. Today $\mathcal{H}_0 = H_0/c \simeq 3.3 \times 10^{-4} h \text{ Mpc}^{-1}$. How large is the neglected term at $k = 0.1 h \text{ Mpc}^{-1}$, and at $k = 10^{-3} h \text{ Mpc}^{-1}$? Which of the two lies inside the window quoted in Box 12? (iii) The assumption in (i) can be removed rather than justified: combining the time–time constraint with the time–space one gives an algebraic equation for ϕ carrying no time derivative at all, in which the correction is manifestly $\mathcal{O}(\mathcal{H}/k)$ [9, Ex. 6.7]. Why is eliminating an assumption preferable to bounding it?

Exercise 1.4. Growth by hand. In Einstein–de Sitter ($\Omega_m = 1$), $a \propto \tau^2$ and $\mathcal{H} = 2/\tau$, so (1.6) reads $\ddot{D} + (2/\tau)\dot{D} - (6/\tau^2)D = 0$. (i) Try $D \propto \tau^n$ and factorize the indicial equation. (ii) Identify which root is the growing mode, and express both in terms of a rather than τ . (iii) A second-order equation has two solutions and the general one mixes them. Relative to the growing mode, how fast does the other die, and what exactly does that erase?

Exercise 1.5. The Mészáros effect. Take a universe of cold dark matter and radiation, with the radiation smooth — it does not cluster — but dominating the expansion until equality. Using $y \equiv a/a_{\text{eq}}$ as the time variable, the CDM density contrast obeys

$$\frac{d^2\delta}{dy^2} + \frac{2+3y}{2y(1+y)} \frac{d\delta}{dy} - \frac{3}{2y(1+y)} \delta = 0.$$

(i) Verify that $\delta \propto 1 + \frac{3}{2}y$ solves it. (ii) Show that this is frozen for $y \ll 1$ and grows as a for $y \gg 1$: a mode sitting inside the horizon during radiation domination gains almost nothing until equality. (iii) Section 1.5 got the suppression of $T(k)$ from the potential decaying while a mode waits. Here it comes from the density refusing to grow. Are these two effects or one, and which is easier to turn into a number?

Lecture 2: From the power spectrum to the cosmic web

2.1 The Lagrangian picture

How does gravitational collapse actually proceed? A slightly overdense region decelerates the matter around it; particles drift toward it, overshoot, and eventually orbit. The natural variable is not the density at a fixed point but the trajectory of each mass element. Label each element by its initial comoving position $\mathbf{q}$ and track its displacement,

$$\mathbf{x}(\mathbf{q}, \tau) = \mathbf{q} + \mathbf{\Psi}(\mathbf{q}, \tau), \quad \mathbf{\Psi}(\mathbf{q}, 0) = 0. \quad (2.1)$$

This is Lagrangian perturbation theory (LPT) [2]. The particle equation of motion (1.3) governs $\mathbf{\Psi}$; taking its divergence and using the Poisson equation,

$$\nabla_{\mathbf{x}} \cdot [\ddot{\mathbf{\Psi}} + \mathcal{H} \dot{\mathbf{\Psi}}] = -\frac{3}{2} \Omega_m(\tau) \mathcal{H}^2 \delta(\mathbf{x}(\mathbf{q}, \tau)). \quad (2.2)$$

The density needs no equation of its own. Mass conservation, $\bar{\rho} d^3q = \rho(\mathbf{x}) d^3x$, fixes it algebraically from the volume distortion of the map:

$$1 + \delta(\mathbf{x}, \tau) = \frac{1}{J(\mathbf{q}, \tau)}, \quad J = \det(\delta_{ij} + \Psi_{i,j}), \quad \Psi_{i,j} \equiv \frac{\partial \Psi_i}{\partial q_j}. \quad (2.3)$$

A cell that shrinks ($J < 1$) is overdense, one that expands is underdense; Figure 4 shows both. Because the determinant is nonlinear in $\mathbf{\Psi}$, a first-order displacement already generates δ at every order: LPT resums the Eulerian nonlinearities that come from moving matter. Only those, though. The Jacobian gives every power of the linear displacement, not every dynamical nonlinearity, so more orders in δ is not more dynamical accuracy.

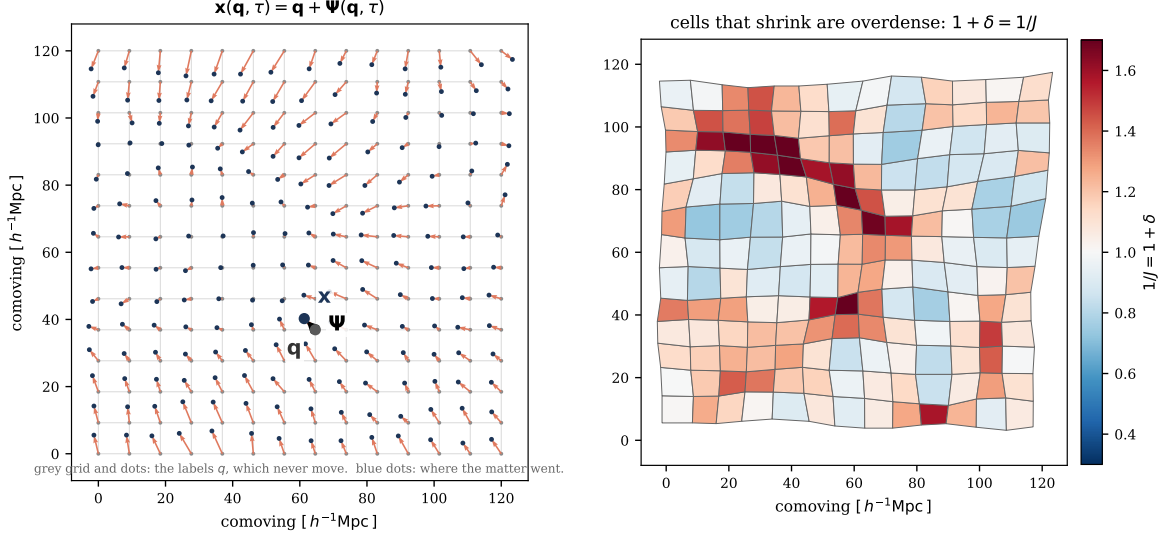


Figure 4: The Lagrangian map and its Jacobian. *Left:* a regular grid of labels $\mathbf{q}$, with an arrow to where the Zel'dovich displacement sends each one. The labels are bookkeeping and never move; the matter does. *Right:* the same cells drawn as the quadrilaterals they become, shaded by their area ratio, which in two dimensions is $1/J = 1 + \delta$. Cells that shrink go red, and they line up into a ridge, which is a filament beginning. The field is the linear spectrum of Lecture 1 projected to two dimensions, with the displacement amplitude set to 0.46 of a cell so that the distortion is visible; a three-dimensional spectrum on a two-dimensional field has no meaningful normalization of its own.

Take (2.2) to first order, where ∇_x and ∇_q agree because they differ at order $\boldsymbol{\Psi}$. As a first-order approximation the determinant in (2.3) is $J \simeq 1 + \nabla_q \cdot \boldsymbol{\Psi}^{(1)}$, and since $1 + \delta = J^{-1} \simeq 1 - \nabla_q \cdot \boldsymbol{\Psi}^{(1)}$, the linear solution is simply

$$\nabla_q \cdot \boldsymbol{\Psi}^{(1)}(\mathbf{q}, \tau) = -\delta^{(1)}(\mathbf{q}, \tau) \quad \Longleftrightarrow \quad \boldsymbol{\Psi}^{(1)}(\mathbf{k}, \tau) = \frac{i\mathbf{k}}{k^2} \delta^{(1)}(\mathbf{k}, \tau), \quad (2.4)$$

where the second form uses the irrotational condition $\nabla_q \times \boldsymbol{\Psi}^{(1)} = 0$ of Section 1.2 to introduce a Lagrangian potential, $\boldsymbol{\Psi}^{(1)} = -\nabla_q \varphi^{(1)}$.

Mass conservation alone fixes (2.4). The dynamics enters only through the time dependence: substituting $\delta^{(1)} = -\nabla_q \cdot \boldsymbol{\Psi}^{(1)}$ back into (2.2) returns the growth equation (1.6) for $S \equiv \nabla_q \cdot \boldsymbol{\Psi}^{(1)}$,

$$\ddot{S} + \mathcal{H}\dot{S} - \frac{3}{2}\Omega_m\mathcal{H}^2S = 0,$$

so we may factor out the same linear growth factor as in Lecture 1, $\boldsymbol{\Psi}^{(1)}(\mathbf{q}, \tau) \propto D_+(\tau)$. This is why no $\Omega_m\mathcal{H}^2$ appears in (2.4): the cosmology of (2.2) sits entirely in D_+ , which Section 2.2 factors out of the eigenvalues.

2.2 The Zel'dovich approximation

The *Zel'dovich approximation* (ZA) [3] takes this linear displacement seriously beyond linear order in the density: move particles on straight lines (in comoving coordinates) and read the density off the Jacobian.

Since $\boldsymbol{\Psi}^{(1)} = -\nabla_q \varphi^{(1)}$, the deformation matrix $\Psi_{i,j}^{(1)} = -\partial^2 \varphi^{(1)} / \partial q_i \partial q_j$ is a Hessian and therefore symmetric, which is what gives it three real eigenvalues and three orthogonal eigenvectors. At each $\mathbf{q}$ there are then three perpendicular axes along which the displacement only stretches or compresses. Write the eigenvalues as $-D_+ \lambda_i$, where D_+ factors out because $\boldsymbol{\Psi}^{(1)} \propto D_+$, leaving

the λ_i fixed by the initial conditions, and the sign is chosen so that positive λ_i means collapse. In that frame $\delta_{ij} + \Psi_{i,j}^{(1)}$ is diagonal with entries $1 - D_+ \lambda_i$, so with $1 + \delta = 1/J$ from (2.3),

$$J = \prod_{i=1}^3 (1 - D_+ \lambda_i) \implies \delta_{\text{ZA}}(\mathbf{x}, \tau) = \frac{1}{(1 - D_+ \lambda_1)(1 - D_+ \lambda_2)(1 - D_+ \lambda_3)} - 1. \quad (2.5)$$

Each axis contributes one factor, and each factor can vanish on its own. That is the whole of Section 2.3.

The recipe is completely explicit, and every initial-condition generator uses it. Draw $\delta^{(1)}(\mathbf{k})$ with variance $P_L(k)$, multiply by $i\mathbf{k}/k^2$ to get $\Psi^{(1)}$, transform back, and move each particle from $\mathbf{q}$ to $\mathbf{q} + \Psi^{(1)}$. Three Fourier transforms and a shift, and you have a nonlinear density field.

The velocities come free. Since $\Psi^{(1)} \propto D_+$ and nothing else in it depends on time,

$$\mathbf{u}_{\text{ZA}} = \dot{\Psi}^{(1)} = \frac{\dot{D}_+}{D_+} \Psi^{(1)} = \mathcal{H}f \Psi^{(1)}, \quad (2.6)$$

so positions and velocities are the same field up to the factor $\mathcal{H}f$ of (1.7). A simulation needs both, and the ZA supplies them together, one of the reasons it, and its second-order extension in Section 2.5, are what every code uses to start.

The ZA is not a degraded linear theory. The difference is what each one approximates. Linear theory says the *density* is small. The ZA instead says the *displacement* is well approximated at first order, and then treats the consequences exactly, through a determinant that contains every power of $\Psi^{(1)}$. Since bulk flows are the dominant motion on quasi-linear scales, that is the right thing to get right. It is why the ZA follows matter into sheets, where linear theory only says δ is growing, and why it remains a serviceable description well past the point where $\delta = 1$.

2.3 Why a web, and not a collection of spheres

You have already met a model of gravitational collapse: a uniform overdense sphere, which contracts, turns around and collapses to a point. It is the standard toy problem, it is analytically solvable, and it is the basis of most of what is said about halo formation.

It also essentially never happens.

Look at what a sphere requires in (2.5): all three factors must vanish together, $\lambda_1 = \lambda_2 = \lambda_3$. But $\Psi_{i,j}^{(1)}$ is the second-derivative matrix of a Gaussian random field, that is, a random symmetric 3×3 matrix, and degenerate eigenvalues among such matrices form a set of measure zero. Sample four hundred thousand of them and you will not find one. *Exactly spherical collapse has probability zero.*

So the generic situation is $\lambda_1 > \lambda_2 > \lambda_3$, strictly, and the three factors in (2.5) vanish at three different times. That single fact is the origin of the cosmic web. Collapse runs along the λ_1 axis first and the patch flattens into a *sheet*, a Zel'dovich pancake. If $\lambda_2 > 0$ the sheet later contracts along a second axis into a *filament*. If $\lambda_3 > 0$ as well, the filament finally draws up into a *knot*. How far a patch gets is decided by nothing more than how many of its eigenvalues are positive, and that is fixed in the initial conditions, before any dynamics has happened.

That makes the morphology countable. For a Gaussian field the joint distribution of the three eigenvalues is known [7], and the volume fractions come out as

positive eigenvalues	fraction of volume	fate
0	8%	void, expanding in all three directions
1	42%	sheet
2	42%	filament
3	8%	knot

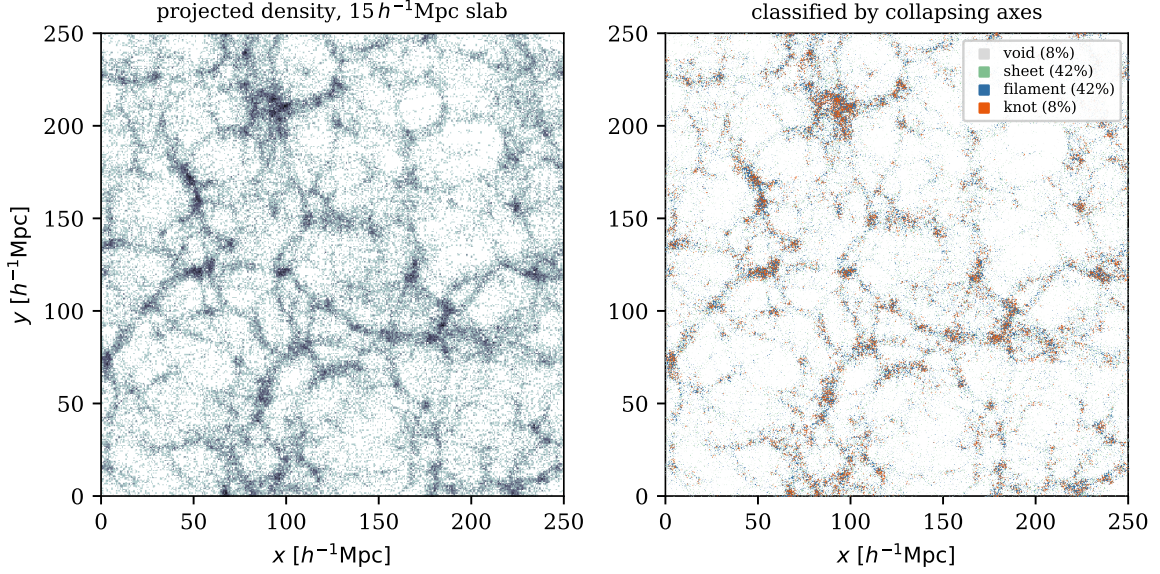


Figure 5: A cosmic web, made by displacing a Gaussian field once. 128^3 particles in a $250 h^{-1}\text{Mpc}$ box at $z = 0$, with the linear spectrum of Lecture 1 and a single Zel’dovich step, with no simulation and no dynamics beyond (2.4). *Left*: projected density through a $15 h^{-1}\text{Mpc}$ slab. *Right*: every particle coloured by how many eigenvalues of its deformation tensor are collapsing. The measured volume fractions are 8.0, 42.1, 42.0 and 8.0 per cent, reproducing the table above from an actual realization rather than from theory. Knots land on the nodes of the web and filaments trace the strands, which is the claim of this section made visible. This is what the hands-on session builds, using the same conventions as the Cosmic Web Sandbox [5].

Compare that with three fair coin flips, which would give 12.5/37.5/37.5/12.5 per cent. The measured split keeps the symmetry, since a Gaussian field prefers no sign of δ , but shifts weight towards the mixed cases: the eigenvalues are not independent, they repel. Read the middle rows: **84% of the volume is headed for a sheet or a filament, and only 8% ever collapses along all three axes.** The web is not decoration draped over a population of spherical halos. It is what almost everything becomes, and the compact objects are the rare case.

This is also why the ZA is worth more than its accuracy suggests. It predicts the skeleton of the cosmic web, where the sheets and filaments will be, from the second derivatives of the initial potential alone, with no dynamics whatsoever. The hands-on session that follows this lecture puts that to the test: you will displace a Gaussian field once, classify each particle by how many of its eigenvalues are collapsing, and get a web out.

2.4 Shell crossing, and where the approximation fails

The first of those collapses is also the end of the theory. When $D_+\lambda_1 \rightarrow 1$ the first factor in (2.5) vanishes, the density formally diverges, and $J \rightarrow 0$. That moment is *shell crossing*: particle trajectories intersect, the map $\mathbf{q} \mapsto \mathbf{x}$ stops being invertible, and several streams with different velocities occupy the same point. The fluid description of Section 1.1 assumed exactly one velocity at each position, which is what A2 meant by cold matter, so it ends here, and the Zel’dovich solution ends with it.

Formally the second factor vanishes later wherever $\lambda_2 > 0$ and the third later still wherever $\lambda_3 > 0$, and the sequence sheet $\rightarrow$ filament $\rightarrow$ knot is that ordering of eigenvalue crossings. As a guide to *morphology* it is excellent, and it is what the table above counts. As *dynamics* past the first caustic it is an extrapolation: a Zel’dovich knot is not a halo, and the approximation says nothing about binding, virialization, or the stable orbiting configuration that collapsed matter

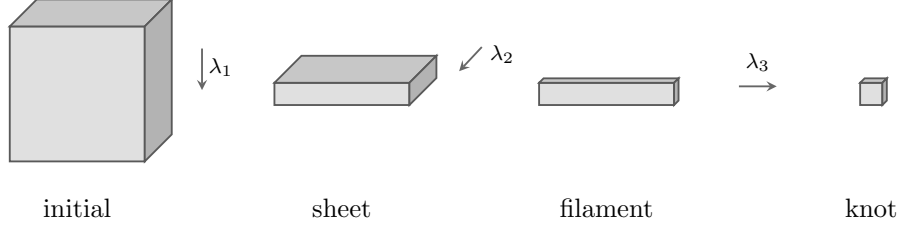


Figure 6: The Zel'dovich eigenvalue sequence, drawn in oblique projection so that all three axes are visible. A patch flattens along its principal axes in order of decreasing λ_i , each stage occurring when the corresponding factor $1 - D_+ \lambda_i$ in (2.5) passes through zero: first the vertical axis, giving a sheet, then the depth axis, then the last one. The second and third stages occur only where λ_2 and λ_3 are themselves positive, so many patches stop at a sheet. Only the first crossing is a genuine ZA prediction, and even that requires a sufficiently positive λ_1 and enough subsequent growth, so it is not guaranteed at an arbitrary point in a late-time accelerating universe. The second and third stages lie beyond the first caustic, where the approximation has already broken down, and they track the morphology of the cosmic web rather than describing collapse. The final object is a knot in the deformation map, not a virialized halo.

actually settles into. Getting through shell crossing needs adhesion models, phase-space methods, or N -body simulation, which is precisely why the initial conditions of Section 2.5 matter, since a simulation is what you hand them to.

2.5 Second order: 2LPT and initial conditions

At second order the displacement responds to the gravitational tide generated at first order. Expanding (2.2) and (2.3) to second order, the divergence $\Psi_{i,i}^{(2)}$ obeys the linear-growth operator with a quadratic source [12, Eqs. (E.30)–(E.31)]; separating variables, the second-order growth factor obeys

$$\ddot{D}_2 + \mathcal{H}\dot{D}_2 - \frac{3}{2}\Omega_m\mathcal{H}^2 D_2 = -\frac{3}{2}\Omega_m\mathcal{H}^2 D_+^2, \quad D_2(\tau) \simeq -\frac{3}{7}D_+^2(\tau)\Omega_m^{-1/143}, \quad (2.7)$$

with the fit better than 0.1% over $10^{-3} < a < 1$ for flat Λ CDM, and $D_2 = -\frac{3}{7}D_+^2$ in EdS. (This Lagrangian D_2 is not the second-order amplitude of Eulerian perturbation theory [8], though the literature uses one symbol for both.) The spatial source is the second principal invariant of $\varphi_{,ij}^{(1)}$; writing $\Psi^{(2)} = \nabla_q \varphi^{(2)}$,

$$\nabla_q^2 \varphi^{(2)} \simeq -\frac{3}{7}\Omega_m^{-1/143} \sum_{i>j} \left[\varphi_{,ii}^{(1)} \varphi_{,jj}^{(1)} - (\varphi_{,ij}^{(1)})^2 \right], \quad (2.8)$$

with $\Psi^{(1)} = -\nabla_q \varphi^{(1)}$ and $\nabla_q^2 \varphi^{(1)} = \delta^{(1)}$, both understood as full-time fields with their growth factors included. Do not read (2.8) as a purely tidal source: the invariant decomposes into an isotropic piece built from δ^2 and a trace-free tidal piece, and it is nonzero even for a spherical patch, as Exercise 2.1 makes explicit. The full 2LPT map, including velocities, is

$$\mathbf{x} = \mathbf{q} - \nabla_q \varphi^{(1)} + \nabla_q \varphi^{(2)}, \quad \mathbf{u} = -\mathcal{H}f_1 \nabla_q \varphi^{(1)} + \mathcal{H}f_2 \nabla_q \varphi^{(2)}, \quad f_2 \simeq 2\Omega_m^{6/11}, \quad (2.9)$$

where $f_1 \equiv d \ln D_+ / d \ln a = f$ and, because the convention here has $D_2 < 0$ so that $\ln D_2$ is undefined,

$$f_2 \equiv \frac{d \ln |D_2|}{d \ln a} = \frac{\dot{D}_2}{\mathcal{H}D_2}. \quad (2.10)$$

The velocity is unaffected, since the sign of D_2 already sits in $\varphi^{(2)}$. Physically: Zel'dovich streams are straight, and the tide bends them. 2LPT is the standard for N -body initial conditions because

starting from the ZA alone leaves slowly decaying transients; the residual depends on starting redshift, force resolution and which statistic you measure [12, App. E.5].

Box 13. Sign conventions. Everything in this lecture (the displacement $\mathbf{x} = \mathbf{q} + \Psi$, the Jacobian, the deformation tensor and its eigenvalues, the growth equation for D_2 and its fit, the second-order source, and the map (2.9) itself) follows Jeong [12, App. E] exactly, including $D_2 < 0$ with $+\nabla_q \varphi^{(2)}$ in the map. Formulae carry across in either direction without adjustment. Other references absorb the minus sign into D_2 instead; check before comparing codes.

Do the Lagrangian and Eulerian descriptions carry the same information? Order by order, and before shell crossing, yes: the companion notes [8] make the correspondence explicit by recovering the Eulerian kernels from the displacement map.

2.6 Galaxies: linear bias

Everything so far has been about matter. Surveys measure galaxies, and galaxies are not matter.

They live in *halos*, the collapsed and virialized lumps of dark matter that Section 2.4 said the Zel'dovich approximation cannot make, and halos sample the density field unfairly. On large scales,

$$\delta_g(\mathbf{x}) = b_1 \delta(\mathbf{x}), \quad \implies \quad P_g(k) = b_1^2 P_L(k), \quad (2.11)$$

with a single number b_1 , the *linear bias*. Rather than assert $b_1 > 1$, we can derive it, in a model crude enough to do on the board and sharp enough to get the mechanism right [6, §2.1].

Halos as regions above a threshold. Smooth the same field $\delta^{(1)}(\mathbf{q})$ this lecture has been displacing on a scale R , and declare a halo wherever it exceeds a fixed δ_{cr} . That is the entire model: no dynamics, one number, the same initial field as Section 2.3 sorted by a different criterion. The number is $\delta_{\text{cr}} \simeq 1.686$, the linearly extrapolated density at which a uniform sphere would collapse to a point.

Box 14. Spherical collapse and δ_{cr} . A threshold taken from spherical collapse is still allowed, because the two statements are about different things. Section 2.3 is about *morphology*: exact degeneracy of the three eigenvalues has probability zero, so nothing collapses as a sphere and the web is not a population of balls. Here δ_{cr} is used only as a *number*, a rough scale for how deep a well must be before it holds together. Nothing in the derivation of (2.13) requires the collapse to be spherical. A different threshold shifts ν_c and the value of b_1 and changes nothing about the mechanism, which is one reason this model predicts a mechanism rather than a number.

Since $\delta^{(1)}$ is Gaussian with variance $\sigma^2(R)$, the fraction of space above the cut is

$$\bar{n} = \frac{1}{2} \operatorname{erfc}\left(\frac{\nu_c}{\sqrt{2}}\right), \quad \nu_c \equiv \frac{\delta_{\text{cr}}}{\sigma(R)}, \quad (2.12)$$

so massive halos, which have large R and hence small $\sigma(R)$ and large ν_c , are exponentially rare. And $\sigma(R)$ is nothing new: σ_8 is this same quantity at $R = 8 h^{-1} \text{Mpc}$.

Why the count is biased. Add a long mode δ_L , constant across a patch many halos wide. Lifting the field is the same as lowering the cut, to $\delta_{\text{cr}} - \delta_L$, so differentiate the abundance:

$$b_1^L = \frac{1}{\bar{n}} \frac{\partial \bar{n}}{\partial \delta_L} = \sqrt{\frac{2}{\pi}} \frac{e^{-\nu_c^2/2}}{\sigma(R) \operatorname{erfc}(\nu_c/\sqrt{2})} \xrightarrow{\nu_c \gg 1} \frac{\delta_{\text{cr}}}{\sigma^2(R)}, \quad (2.13)$$

and there is the answer. Rarer objects sit further out in the tail and respond harder, which is why the most massive galaxies are the most strongly biased, and why b_1 belongs to the tracer, not to the cosmology. Figure 7 is (2.13) measured rather than illustrated.

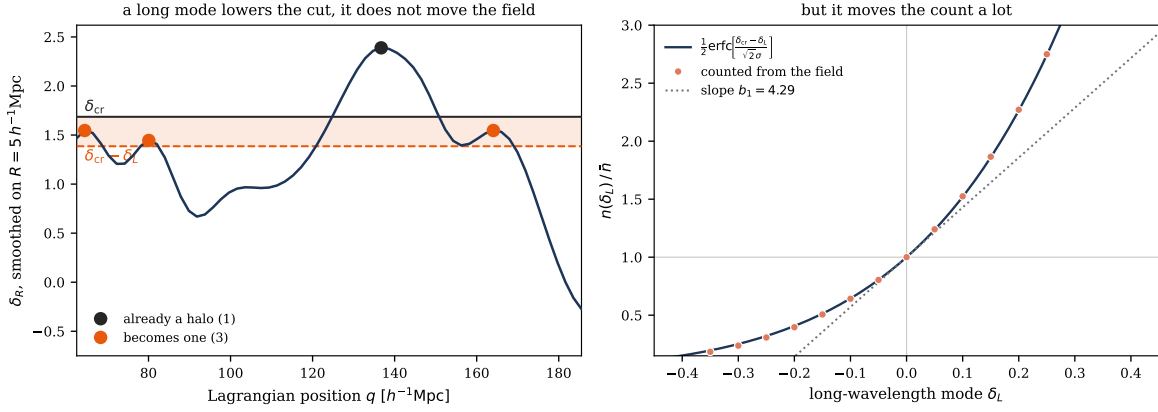


Figure 7: Bias, measured from the same linear field this lecture displaces. *Left, the mechanism.* A short cut through $\delta^{(1)}$ smoothed on $R = 5 h^{-1} \text{Mpc}$, where $\sigma(R) = 0.665$ and $\nu_c = 2.54$. Adding a long mode δ_L to the field is the same as lowering the cut to $\delta_{\text{cr}} - \delta_L$, which is the step (2.13) takes, so the figure draws one field and two thresholds. One peak here was already above; three more sit in the band and become halos when the long mode arrives. This panel shows the mechanism, not a fair sample, and it cannot be one. At this rarity the field crosses δ_{cr} about once per $1600 h^{-1} \text{Mpc}$, so an honest cut long enough to show a *count* changing would run three thousand smoothing lengths and every peak in it would be invisible. The counting is the right panel's job. *Right:* the count against the long mode. Points are counted from the field, the curve is (2.12) with the threshold shifted, and the dotted line is its slope at the origin, (2.13). Measuring that slope from the counts returns 4.31 against a predicted 4.29. Two cautions the picture makes plain: the curve is convex, so a slope fitted across a wide range overestimates b_1 (5.1 here), and the asymptotic form $\delta_{\text{cr}}/\sigma^2 = 3.81$ is still 11% low at this rarity.

Two limitations of (2.11). It is the first term of an expansion, not a law, and the threshold model is far too crude to predict a real b_1 , which is measured alongside the cosmology. At smaller scales the response stops being linear, picks up a dependence on the tidal field, and acquires a stochastic piece. The leading stochastic term is shot noise: galaxies are counted rather than smooth, so each one pairs with itself and adds $\delta_D(\mathbf{r})/n_g$ to the correlation function, hence a constant $1/n_g$ to $P_g(k)$ at every k , with n_g the mean comoving number density. It is not a small-scale effect in origin, only in consequence: it comes from $\mathbf{r} = 0$, and it dominates wherever P_L has fallen below it. And it is written in real space, whereas a survey infers radial positions from redshifts, so peculiar velocities distort what is measured along the line of sight. Both corrections, and the one-loop machinery that organises them, are the subject of the companion notes [8].

Section 1.5 showed how to get $P_L(k)$ from a cosmological model, and it has since been put to two distinct uses. Fed through the displacement map of this lecture it becomes initial conditions for an N -body simulation, which is what the hands-on session does. Multiplied by b_1^2 it becomes a prediction for what a galaxy survey measures. The first use leads to simulations of structure formation, the second to cosmological inference from galaxy clustering.

Exercise 2.1. Solve (2.7) in EdS ($\mathcal{H} = 2/\tau$, $\Omega_m = 1$, $D_+ = a \propto \tau^2$) and verify $D_2 = -\frac{3}{7}a^2$.

Exercise 2.2. (Optional.) In one spatial dimension the Zel'dovich approximation is exact until shell crossing. Prove this by showing that the 1D displacement $\Psi^{(1)}$ solves the full nonlinear equation of motion, using the fact that in 1D the force on a sheet depends only on the mass on either side of it. Assume plane-parallel sheets, a subtracted homogeneous background, and boundary conditions (periodic, or a fixed reference point) that determine the constant in $\int \delta dx$.

Appendices

A short glossary of words the lectures use without ceremony.

A Glossary

Words used in the lectures without ceremony, in case any are new.

Boltzmann code	program integrating the linear relativistic equations for all species (CAMB, CLASS); supplies P_L and growth quantities
$\mathcal{R}$	the comoving curvature perturbation, the quantity inflation hands over; conserved outside the horizon in any background, which is what makes it the anchor. Often written ζ
transfer function	the resulting ratio of a mode's late-time to primordial amplitude, encoding processing through horizon entry and equality
Mészáros effect	the near-freezing of CDM growth for modes inside the horizon during radiation domination
h	the Hubble constant in units of $100 \text{ km s}^{-1} \text{ Mpc}^{-1}$; distances carry $h^{-1} \text{ Mpc}$ so that this choice cancels
N -body simulation	direct numerical integration of gravitating particles, valid past shell crossing where perturbation theory is not
tracer	any observed population standing in for the matter field (galaxies, quasars, halos)

References

- [1] C.-P. Ma and E. Bertschinger, *Cosmological perturbation theory in the synchronous and conformal Newtonian gauges*, *Astrophys. J.* **455** (1995) 7 [[astro-ph/9506072](#)].
- [2] F. Bernardeau, S. Colombi, E. Gaztañaga and R. Scoccimarro, *Large-scale structure of the universe and cosmological perturbation theory*, *Phys. Rept.* **367** (2002) 1 [[astro-ph/0112551](#)].
- [3] Y. B. Zel'dovich, *Gravitational instability: An approximate theory for large density perturbations*, *Astron. Astrophys.* **5** (1970) 84.
- [4] D. J. Eisenstein and W. Hu, *Baryonic features in the matter transfer function*, *Astrophys. J.* **496** (1998) 605 [[astro-ph/9709112](#)].
- [5] N.-M. Nguyen, *The Cosmic Web Sandbox*, [minhmpa.github.io/lss-lab/cosmic-web-sandbox](#).
- [6] V. Desjacques, D. Jeong and F. Schmidt, *Large-scale galaxy bias*, *Phys. Rept.* **733** (2018) 1 [[1611.09787](#)].
- [7] A. G. Doroshkevich, *Spatial structure of perturbations and origin of galactic rotation in fluctuation theory*, *Astrophysics* **6** (1970) 320 [translated from *Astrofizika* **6** (1970) 581].
- [8] N.-M. Nguyen, *Cosmological perturbation theory from matter to galaxies*, companion lecture notes, in preparation.
- [9] S. Dodelson and F. Schmidt, *Modern Cosmology*, 2nd edition, Academic Press (2021).
- [10] D. Huterer, *Growth of cosmic structure*, *Astron. Astrophys. Rev.* **31** (2023) 2 [[2212.05003](#)].
- [11] E. V. Linder, *Cosmic growth history and expansion history*, *Phys. Rev. D* **72** (2005) 043529 [[astro-ph/0507263](#)].
- [12] D. Jeong, *Cosmology with high ($z > 1$) redshift galaxy surveys*, PhD dissertation, University of Texas at Austin (2010).